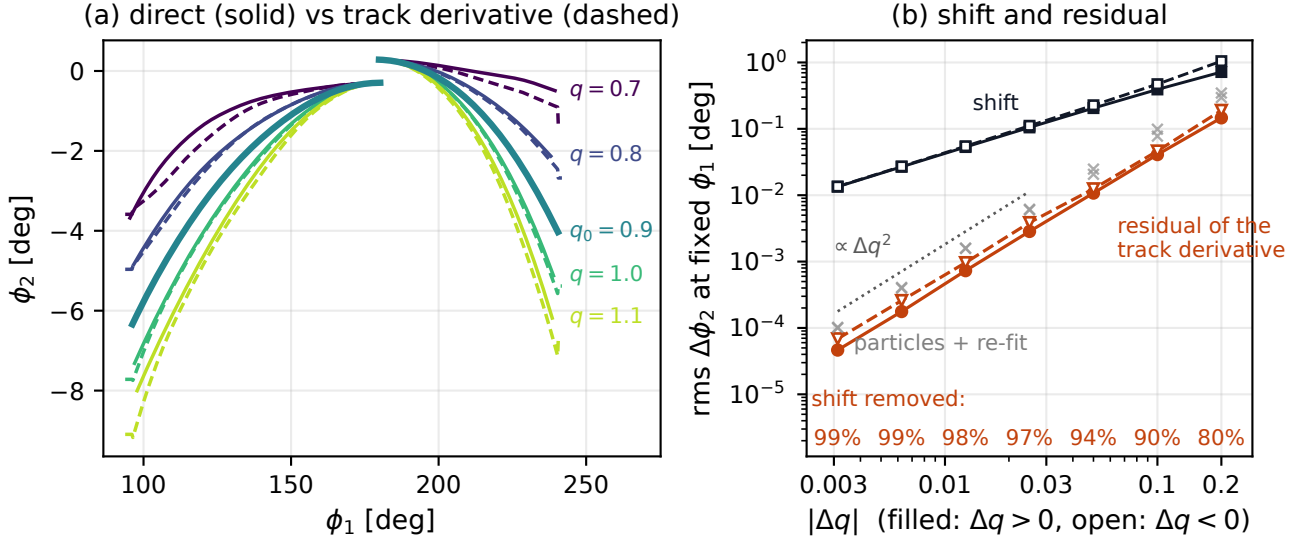
(a) direct (solid) vs track derivative (dashed)
ϕ_2 [deg]
ϕ_1 [deg]
0
−2
−4
−6
−8
100
150
200
250
$q = 0.7$
$q = 0.8$
$q_0 = 0.9$
$q = 1.0$
$q = 1.1$
(b) shift and residual
rms $\Delta\phi_2$ at fixed ϕ_1 [deg]
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
shift
$\propto \Delta q^2$
residual of the
track derivative
particles + re-fit
shift removed:
99% 99% 98% 97% 94% 90% 80%
0.003
0.01
0.03
0.1
0.2
$|\Delta q|$ (filled: $\Delta q > 0$, open: $\Delta q < 0$)